Special topics in logic and security II
Exam solutions — May 21, 2026

Problem 1 — A control matrix for $\mathrm{Ham}_3(3)$

By definition (codingth.pdf, §4), $\mathrm{Ham}_q(k)$ is the linear code over $\mathbb{F}_q$ whose check matrix has as columns the $n = \frac{q^k-1}{q-1}$ representatives of the projective points in $\mathbb{P}^{k-1}(\mathbb{F}_q)$.

For $q = k = 3$ we get $n = \frac{3^3-1}{3-1} = 13$. We list the 13 projective points of $\mathbb{P}^2(\mathbb{F}_3)$ in normalised form (first non-zero coordinate equal to 1) and place them as columns of $H \in M_{3\times 13}(\mathbb{F}_3)$:

$$H = \left(\begin{array}{ccccccccc|ccc|c} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 & 2 & 2 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 1 & 2 & 1 \end{array}\right).$$

Since $\operatorname{rk} H = 3$ (the first, tenth and thirteenth columns are e_1, e_2, e_3), H is indeed a valid check (= control) matrix of the $[13, 10, 3]$ code $\mathrm{Ham}_3(3)$. Any two columns of H are linearly independent (distinct projective points), but $\langle h_1\rangle, \langle h_2\rangle, \langle h_1 + h_2\rangle$ are linearly dependent, so $d(\mathrm{Ham}_3(3)) = 3$.

Problem 2 — Weight polynomial of $\mathrm{Sim}_3(3)$

The simplex code $\mathrm{Sim}_q(k)$ has the generating matrix equal to the check matrix of $\mathrm{Ham}_q(k)$, hence parameters $\left[\frac{q^k-1}{q-1},\, k,\, q^{k-1}\right]$. The theorem in §4 of codingth.pdf states that *every* nonzero codeword of $\mathrm{Sim}_q(k)$ has the same weight q^{k-1}.

For $\mathrm{Sim}_3(3)$: $n = 13$, $k = 3$, every nonzero word has weight $3^2 = 9$, and $|\mathrm{Sim}_3(3)| = 3^3 = 27$. Therefore

$$\boxed{A_{\mathrm{Sim}_3(3)}(z) \;=\; 1 \;+\; 26\, z^9}$$

Problem 3 — Weight polynomial of $\mathrm{Ham}_3(3)$ via MacWilliams

Because $\mathrm{Sim}_q(k) = \mathrm{Ham}_q(k)^\perp$, MacWilliams' formula (codingth.pdf, Thm. 14.2) gives

$$A_{\mathrm{Ham}_3(3)}(z) \;=\; q^{-k}\,(1 + (q-1)z)^n\, A_{\mathrm{Sim}_3(3)}\!\left(\frac{1-z}{1+(q-1)z}\right)$$

with $q = 3$, $n = 13$, $k = 3$ (the dimension of the inner code $\mathrm{Sim}_3(3)$):

$$A_{\mathrm{Ham}_3(3)}(z) \;=\; \frac{1}{27}(1+2z)^{13}\left[1 + 26\left(\frac{1-z}{1+2z}\right)^9\right] \;=\; \frac{(1+2z)^{13} + 26\,(1-z)^9(1+2z)^4}{27}.$$

Expansion. Computing the two summands:
$(1+2z)^{13} = \sum_{k=0}^{13} \binom{13}{k} 2^k z^k$ has coefficients

$$[1,\ 26,\ 312,\ 2288,\ 11440,\ 41184,\ 109824,\ 219648,\ 329472,\ 366080,\ 292864,\ 159744,\ 53248,\ 8192].$$

$(1-z)^9(1+2z)^4$ has coefficients

$$[1,\ -1,\ -12,\ 20,\ 46,\ -126,\ -12,\ 300,\ -279,\ -121,\ 400,\ -312,\ 112,\ -16].$$

Multiplying the second row by 26, adding the two rows term-by-term and dividing by 27 yields the weight distribution A_i of $\mathrm{Ham}_3(3)$:

i	A_i
0	1
1	0
2	0
3	104
4	468
5	1404
6	4056
7	8424

i	A_i
8	11934
9	13442
10	11232
11	5616
12	2080
13	288
$\sum_i A_i = 59049 = 3^{10}$	

Hence

$$\boxed{\begin{aligned} A_{\mathrm{Ham}_3(3)}(z) &= 1 + 104\,z^3 + 468\,z^4 + 1404\,z^5 + 4056\,z^6 + 8424\,z^7 + 11934\,z^8 \\ &\quad + 13442\,z^9 + 11232\,z^{10} + 5616\,z^{11} + 2080\,z^{12} + 288\,z^{13}. \end{aligned}}$$

Consistency check. $A_{\mathrm{Ham}_3(3)}(1) = 3^{10} = 59049 = |C|$, and $A_1 = A_2 = 0$ as required by $d = 3$. Moreover $A_3 = 104$ can be verified directly by combinatorics in $\mathbb{P}^2(\mathbb{F}_3)$: the projective plane has 13 lines, each containing $q + 1 = 4$ points; collinear unordered triples are $13 \cdot \binom{4}{3} = 52$, and each gives $q - 1 = 2$ weight-3 codewords (the non-zero scalar multiples of the unique dependence), yielding $52 \cdot 2 = 104$.

Problem 4 — $W_2 \otimes W_2$ is unitary on H_4

Recall the Hadamard–Walsh matrix $W_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Using the block definition of the Kronecker product,

$$W_2 \otimes W_2 \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} W_2 & W_2 \\ W_2 & -W_2 \end{pmatrix} \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \;=:\; W_4.$$

Unitarity — tensor argument. W_2 is real symmetric and satisfies $W_2^2 = \frac{1}{2}\left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right) = I_2$, so $W_2^* = W_2 = W_2^{-1}$. Applying the mixed-product property (Problem 5 of the previous exam):

$$(W_2 \otimes W_2)^*(W_2 \otimes W_2) \;=\; (W_2^* \otimes W_2^*)(W_2 \otimes W_2) \;=\; W_2^*W_2 \otimes W_2^*W_2 \;=\; I_2 \otimes I_2 = I_4.$$

Unitarity — direct computation. W_4 is real symmetric, so $W_4^* = W_4$. Compute W_4^2: each row of the ± 1 matrix has Euclidean square-norm 4, and any two distinct rows are orthogonal (each pair of rows differs in exactly two positions, contributing $+1 + 1$ minus $+1 + 1 = 0$). Therefore the 4×4 ± 1 matrix squared equals $4I_4$, and $W_4^2 = \frac{1}{4} \cdot 4I_4 = I_4$. Hence $W_4^* W_4 = I_4$, i.e. W_4 is unitary on H_4.

Problem 5 — Real system characterising $A \in U(2)$

Write $A = \begin{pmatrix} \alpha & \gamma \\ \varepsilon & \eta \end{pmatrix}$ with $\alpha = a + ib$, $\gamma = c + id$, $\varepsilon = e + if$, $\eta = g + ih$ and $a, b, \ldots, h \in \mathbb{R}$. By definition A is unitary iff $A^*A = I_2$, where

$$A^*A \;=\; \begin{pmatrix} |\alpha|^2 + |\varepsilon|^2 & \bar{\alpha}\gamma + \bar{\varepsilon}\eta \\ \overline{\bar{\alpha}\gamma + \bar{\varepsilon}\eta} & |\gamma|^2 + |\eta|^2 \end{pmatrix}.$$

The two diagonal entries give

$$|\alpha|^2 + |\varepsilon|^2 = a^2 + b^2 + e^2 + f^2, \qquad |\gamma|^2 + |\eta|^2 = c^2 + d^2 + g^2 + h^2.$$

The off-diagonal entry is

$$\bar{\alpha}\gamma + \bar{\varepsilon}\eta = (a - ib)(c + id) + (e - if)(g + ih) = \underbrace{(ac + bd + eg + fh)}_{\Re} + i\,\underbrace{(ad - bc + eh - fg)}_{\Im}.$$

Setting $A^*A = I_2$ is equivalent to the four real equations

$$\boxed{\begin{array}{ll} (\mathrm{N}_1) & a^2 + b^2 + e^2 + f^2 = 1, \\ (\mathrm{N}_2) & c^2 + d^2 + g^2 + h^2 = 1, \\ (\mathrm{O}_1) & ac + bd + eg + fh = 0, \\ (\mathrm{O}_2) & ad - bc + eh - fg = 0. \end{array}}$$

(N_1), (N_2) say that the columns of A have unit Hermitian norm; (O_1), (O_2) say that the columns are orthogonal in $\mathbb{C}^2$.

Supplementary 1: $\dim_{\mathbb{R}} U(2)$. The eight real parameters $a, b, \ldots, h$ are subject to the four real equations above, and these are independent at any unitary matrix (the differential of the map $A \mapsto A^*A - I_2$ has constant rank 4 on $U(2)$, since the target space of Hermitian matrices is 4-dimensional over $\mathbb{R}$). Hence $U(2)$ is a smooth real submanifold of $\mathbb{R}^8$ of dimension

$$\dim_{\mathbb{R}} U(2) \;=\; 8 - 4 \;=\; 4,$$

which agrees with the standard formula $\dim_{\mathbb{R}} U(n) = n^2$.

Supplementary 2: A single polynomial equation. Yes. Summing the squares of the residuals of (N_1)–(O_2) gives the real polynomial

$$E(a,b,c,d,e,f,g,h) \;=\; \big(a^2{+}b^2{+}e^2{+}f^2{-}1\big)^2{+}\big(c^2{+}d^2{+}g^2{+}h^2{-}1\big)^2{+}\big(ac{+}bd{+}eg{+}fh\big)^2{+}\big(ad{-}bc{+}eh{-}fg\big)^2.$$

Since E is a sum of squares of real numbers, $E \geq 0$ everywhere on $\mathbb{R}^8$, with $E = 0$ iff each of the four summands vanishes, iff (N_1)–(O_2) all hold, iff A is unitary. So $\{E = 0\} = U(2)$. The example shows that a single real polynomial equation can carve out a positive-codimension variety because the squaring trick converts a system into one equation while preserving the zero locus.

Problem 6 — The parallelogram identity for a qubit

Let $q = a|0\rangle + b|1\rangle$ with $a, b \in \mathbb{C}$, $|a|^2 + |b|^2 = 1$.

Solution 1 — algebraic, via $|z|^2 = z\bar{z}$.

$$\begin{aligned} |a+b|^2 &= (a+b)\overline{(a+b)} = (a+b)(\bar{a}+\bar{b}) = |a|^2 + a\bar{b} + b\bar{a} + |b|^2, \\ |a-b|^2 &= (a-b)(\bar{a}-\bar{b}) = |a|^2 - a\bar{b} - b\bar{a} + |b|^2. \end{aligned}$$

Adding the two lines, the cross terms $\pm(a\bar{b} + b\bar{a})$ cancel and

$$|a+b|^2 + |a-b|^2 \;=\; 2\big(|a|^2 + |b|^2\big) \;=\; 2 \cdot 1 \;=\; 2.$$

Solution 2 — quantum, applying the Hadamard gate W_2.

$$W_2\, q \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}a\\ b\end{pmatrix} \;=\; \frac{a+b}{\sqrt{2}}|0\rangle \;+\; \frac{a-b}{\sqrt{2}}|1\rangle.$$

Because W_2 is unitary it preserves norms, so W_2q is again a unit quantum state. Hence the squared amplitudes sum to 1:

$$\left|\frac{a+b}{\sqrt{2}}\right|^2 + \left|\frac{a-b}{\sqrt{2}}\right|^2 = 1 \iff |a+b|^2 + |a-b|^2 = 2. \qquad \blacksquare$$